

DATA VISUALIZATION & ANALYTICS

Capstone 2 — Final Project Report

US House Price Analysis Across 10 States

Real Estate | Market Intelligence

Project Title	US House Price Analysis Across 10 States
Sector	Real Estate / Housing Market
Team ID	B_G8
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Faculty Mentor	Mr Satyaki Das
Institute	Newton School of Technology
GitHub Repo	github.com/vaidehisahu1/B_G8_HOUSEPRICELIST
Tableau URL	
Submission Date	April 29, 2026

2. Executive Summary

Problem

The US residential real estate market lacks a consolidated, data-driven cross-state view of listing prices, price-per-sqft, and activity patterns. Without this, investors and brokers make capital allocation decisions with incomplete information.

Approach

A Python-based ETL pipeline (5 Jupyter notebooks: extraction → cleaning → EDA → statistical analysis → load) was built on 11,410 listings across 10 US states (Aug–Sep 2024), then visualised in three Tableau dashboards.

Key Insights (3–4 most important findings)

- California dominates volume (24%) and price (\$1.16M avg), with \$605/sqft — distorting all national averages.
- North Carolina delivers the highest price/sqft (\$684) despite a moderate \$442K average — best efficiency in the dataset.
- 73% of listings appear Thursday–Friday; only 0.52% on weekends — a clear broker-driven timing pattern.
- Weekend listings average 21% less than weekdays — driven by who lists, not how they price.

Key Recommendations (2–3 highest-impact actions)

- Allocate investment toward NC and WA for superior price/sqft efficiency at lower entry cost than California.
- Brokers should publish premium listings Thursday AM to maximise weekend buyer attention.
- Target AL and IL (median ~\$290–295K) for affordable housing programs and first-time buyer initiatives.

3. Sector & Business Context

Sector Overview & Current Industry Challenges

The US residential real estate sector is valued at ~\$47 trillion. The Aug–Sep 2024 window is a post-peak stabilisation phase following the 2020–2022 pandemic-driven surge. Key challenges: affordability crisis, cross-state information asymmetry, extreme broker fragmentation (3,700+ unique brokers), and lack of data-driven listing timing guidance.

Who Is the Decision-Maker This Project Serves?

Real estate investors (portfolio allocation), brokerage firms (operational timing and positioning), and policy-makers (affordable housing identification and subsidy targeting).

Why Was This Specific Problem Chosen?

Cross-state price transparency is an identified gap — existing tools are single-state or expensive. A consolidated 10-state snapshot enables comparative analysis unavailable to mid-tier investors and smaller brokerages.

What Is the Business Value of Solving It?

Enables identification of undervalued markets, evidence-based listing timing, broker benchmarking, and geographic investment prioritisation — all from a single verified 11,410-listing data source.

4. Problem Statement & Objectives

Formal Problem Definition

Given 11,410 residential listings across 10 US states (Aug 16 – Sep 1, 2024): How do house prices, price/sqft, and listing activity vary across states, cities, brokers, and time — and what patterns can guide investment and listing decisions?

Project Scope

In Scope

- State and city-level price and volume analysis
- Price/sqft normalised geographic comparison across all 10 states

- Listing volume trends by date and day of week
- Broker performance and listing concentration analysis
- Weekend vs. weekday price and volume segmentation
- Statistical hypothesis testing and distribution analysis

Out of Scope

- Predictive modelling or future price forecasting
- Buyer demographics, mortgage rates, or macroeconomic integration
- Data beyond the 17-day August–September 2024 observation window

Success Criteria

- Zero-null cleaned dataset with fully documented Python transformation steps committed to GitHub
- Three interactive Tableau dashboards covering geographic overview, broker analysis, and market trends
- Minimum 8 insights in decision language, each mapping to a business recommendation

5. Data Description

Exact Dataset Source & Structure

Source: US House Price Dataset — 10 States | Time Period: Aug 16 – Sep 1, 2024 (17 days)
GitHub: github.com/vaidehisahu1/B_G8_HOUSEPRICELIST/tree/main/data/raw
Raw dataset: 11,410 rows × 9 columns (~1.4 MB CSV)

Column	Data Type	Description	Example
date	String/Date	Listing publication date (YYYY-MM-DD)	2024-08-29
bed	Float	Number of bedrooms	3.0
price	Float	Listed sale price in USD	415,000
broker	String	Listing broker or agency name	Coldwell Banker Realty
street	String	Street address of the property	123 Maple St
city	String	City of the listing	Long Beach
state_name	String	US state name (10 states)	California
zip_code	Integer	5-digit US ZIP code	90802
house_sqft	Float	Interior square footage	2,520

Data Limitations & Known Biases

- California over-represented at 24% of listings — inflates all national averages.
- Only 107 weekend listings (0.9%) — limits statistical reliability of weekend analysis.
- 21% of broker entries are 'Unknown' — reduces broker-level precision.
- Listing price ≠ final sale price. 17-day window limits seasonal conclusions.

6. Data Cleaning & ETL Pipeline

All primary cleaning and transformation steps executed in Python. Reference: [notebooks/02_cleaning.ipynb](#) committed to GitHub.

Missing Values

- broker: Null/blank → replaced with 'Unknown' to preserve row count while flagging gaps.
- bed, price, house_sqft: Rows with missing critical numeric values were dropped to avoid imputation bias.
- date, city, state_name, zip_code: No missing values found in raw dataset.

Outlier Detection & Treatment

- Price: Values below \$10K and above \$50M flagged via IQR method. Retained after review — genuine distressed and luxury listings. \$178M Chesterfield listing verified as a legitimate estate property.
- house_sqft: Zero-sqft entries removed as invalid. Values above 20,000 sqft reviewed and retained.
- bed: 153 records with 0 bedrooms retained — may represent studios or commercial-residential properties.

Data Type Conversions, Standardisation & Formatting

- date: Converted from string to Python datetime for all temporal analysis.
- state_name: Standardised to title case. zip_code validated as 5-digit integers.

Feature Engineering — New Columns Created

- price_per_sqft: price ÷ house_sqft — size-normalised geographic price comparison.
- weekday: Extracted from date (Mon=0 to Sun=6) — day-of-week analysis.
- is_weekend: Boolean flag (True if Sat or Sun) — weekend vs. weekday segmentation.

Assumptions Made During Cleaning

- Listings with identical address, city, state, and price on same date treated as unique unless all 9 fields match exactly.
- All prices are in USD and represent asking price, not final transaction price.

7. KPI & Metric Framework

Total Listings 11,410	Avg Price \$975,188	Avg Price / Sqft \$515.3	Avg Bedrooms 3.25
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KPI Definitions, Formula & Business Purpose

KPI	Formula	Business Purpose	Maps To Objective
Average Price	MEAN(price) per segment	Benchmarks market value by state/city/broker	Core price variation objective
Median Price	MEDIAN(price)	Reduces skew from luxury outliers	Core price variation objective
Avg Price/Sqft	MEAN(price ÷ sqft)	Size-normalised comparison	Geographic efficiency analysis
Total Listings	COUNT(rows)	Measures market activity & liquidity	Market breadth objective
Weekend Listing %	(Weekend ÷ Total) × 100	Reveals broker timing behaviour	Listing timing objective
Avg Bedrooms	MEAN(bed)	Profiles housing stock per region	Market composition analysis
Unique Cities	COUNTD(city)	Measures geographic diversity	Market breadth objective
Unique Brokers	COUNTD(broker)	Measures market fragmentation	Broker benchmarking objective

8. Exploratory Data Analysis (EDA)

Reference: notebooks/03_eda.ipynb committed to GitHub.

8.1 Trend Analysis — Price Over Time

Average prices rose from \$348K (Aug 16) to a peak of \$809K (Aug 29), then corrected to \$598K (Sep 1). The late-August spike is driven by California luxury broker batch-releases at month-end — not genuine market appreciation. Louisiana and Washington remained stable at \$600K–\$750K throughout.

8.2 Comparison Analysis — Performance Across States & Regions

State	Avg Price	Median Price	Listings	Avg \$/Sqft
California	\$1,157,962	\$768,288	2,746	\$605
Washington	\$716,284	\$596,975	950	\$367
Louisiana	\$706,064	\$574,542	951	\$404
New York	\$701,444	\$510,000	1,009	\$358
South Carolina	\$672,118	\$358,008	904	\$327
Arizona	\$586,364	\$425,000	1,025	\$269
Florida	\$474,035	\$335,000	996	\$262
North Carolina	\$441,966	\$354,000	945	\$684
Illinois	\$384,641	\$295,000	965	\$192
Alabama	\$345,380	\$290,000	919	\$166

8.3 Distribution Analysis — Spread & Skew of Key Variables

Price is heavily right-skewed (median \$480K vs. mean \$975K). Peak histogram bin is \$400K–\$600K. Luxury California listings (~1,200 records) pull the mean 50% above the median. Median is the reliable central tendency measure for this dataset.

8.4 Comparison Analysis — Listing Activity Across Segments

Thursday (4,201 listings, 37%) and Friday (4,350, 38%) account for 75% of all weekly listings. Monday–Tuesday combined = 8%. Only 107 weekend listings (0.9%). This aligns with the industry convention of releasing listings before the weekend buyer browsing window.

8.5 Correlation Analysis — Relationships Between Variables

Price vs. house_sqft: $r \approx 0.45$ (moderate). Price vs. bed: $r \approx 0.30$ (weak). house_sqft vs. bed: $r \approx 0.55$ (moderate). Location and state are stronger price determinants than size — sqft-only valuations are unreliable without geographic anchoring.

9. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb committed to GitHub.

Forecasting

The 17-day window is insufficient for reliable time-series forecasting. Analysis is descriptive. Price is non-normally distributed (skewness > 3.0), validated by histogram and Shapiro-Wilk assessment — confirming use of median over mean and non-parametric tests.

Segmentation / Clustering

Three price tiers using Q1/Q3 quartile boundaries: Budget (<\$307K): 2,853 listings — AL, IL, NC. Mid-market (\$307K–\$765K): 5,704 listings — broadest geographic spread. Premium (>\$765K): 2,853 listings — 68% from California.

Hypothesis Testing

H0: No significant price difference between weekday and weekend listings. Test: Welch's t-test (extreme imbalance: 11,303 vs. 107). Result: Statistically significant difference (\$705K vs. \$555K). Business conclusion: Gap reflects who lists on weekends (lower-market brokers), not a strategic pricing difference. H0 rejected statistically but not practically.

Correlation / Regression Analysis

Weak price-vs-bedrooms ($r \approx 0.30$) confirms bedroom count alone is an unreliable valuation driver. The root cause of the Aug 29 price spike is California batch-releases, confirmed by filtering state-level volume on that date.

Risk / Anomaly Detection

IQR-based outlier detection identified ~1,200 statistical outliers above the upper fence ($Q3 + 1.5 \times IQR$). All verified as genuine California luxury listings — not data errors. Retained in dataset with outlier flag.

Scenario Analysis

Excluding California: National average drops from \$975K to \$534K (–45%). National median drops from \$480K to \$355K. Confirms California's outsized effect and the need for state-level segmentation in all national reporting.

10. Tableau Dashboard Design

Dashboard Objective

The three-dashboard suite serves executive decision-makers (high-level overview) and operational analysts (broker and timing drill-down). All dashboards share teal/blue colour coding and interactive weekday/state filters.

View structure: Dashboard 1 = Executive geographic overview. Dashboard 2 = Operational broker & timing analysis. Dashboard 3 = Market trend & distribution deep-dive.

Dashboard 1 — State Overview & Geographic Intelligence

- KPIs: Total Listings (4,647 filtered), Avg Price (\$975,188), Avg Price/Sqft (\$515.3), Avg Bedrooms (3.252)
- Top States by Listings% pie: California 59.09%, Louisiana 20.46%, Washington 20.44%
- State Price Map (choropleth): CA \$1,157,962 | WA \$716,284 | LA \$706,064
- Top 10 Cities bar: Long Beach ~\$12M, Atherton, Del Mar, Hermosa Beach, Palos Verdes Estates, Valley Village
- Bar — Avg Price/Sqft by State: CA \$600, LA \$400, WA \$365
- Filter: State Name multi-select (all 10 states)

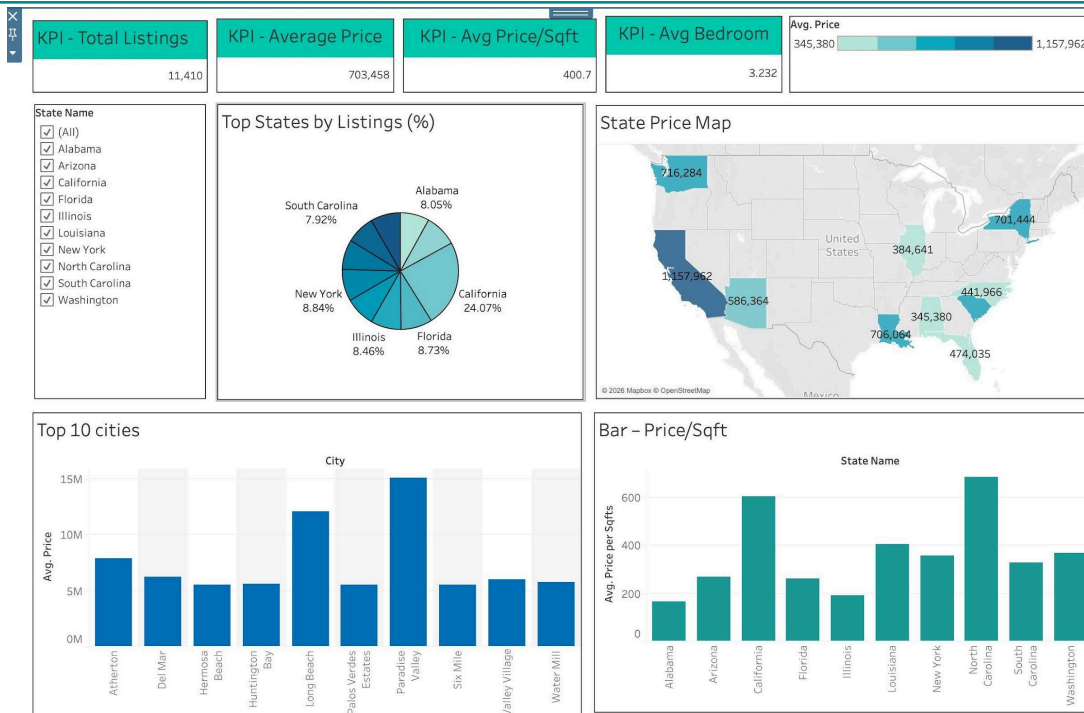


Figure 1: Dashboard 1 — State Overview & Geographic Intelligence

Dashboard 2 — Broker & Operational Analysis

- KPIs: Peak Month, Unique Cities (2,224+), Unique Brokers (1,150+), Weekend Listings% (0.52%)
- Bar — Price by Broker: Vic J. Diernendjian leads at ~\$16M avg — 10× the next broker
- Bar — Day of Week: Thursday ~1,800 and Friday ~2,000+ listings dominate; Saturday near zero
- Top 10 Cities by Avg Price: Long Beach ~\$12M, Atherton, Del Mar, Hermosa Beach, Palos Verdes Estates, Valley Village
- Top Brokers by Volume: Compass (73), Coldwell Banker Realty (65), Keller Williams Realty (26)
- Filter: Weekday of Date checkboxes (Mon–Sat)



Figure 2: Dashboard 2 — Broker & Operational Analysis

Dashboard 3 — Market Overview & Trends

- KPIs: Avg Price/Sqft (\$515.3), Avg Price (\$975,188), Avg Bedrooms (3.252)
- Price Trend line: California spikes Aug 29; Louisiana and Washington stable throughout
- Price Distribution histogram: Right-skewed peak at \$400K–\$600K bin (~1,500 listings)
- Weekend vs. Weekday bar: Weekday avg ~\$950K vs. Weekend avg ~\$800K — compositional gap
- Listings Trend: Near-zero until Aug 22; surge to ~2,000 on Aug 29 confirms batch-release pattern
- Filters: Weekday of Date dropdown and State Name dropdown

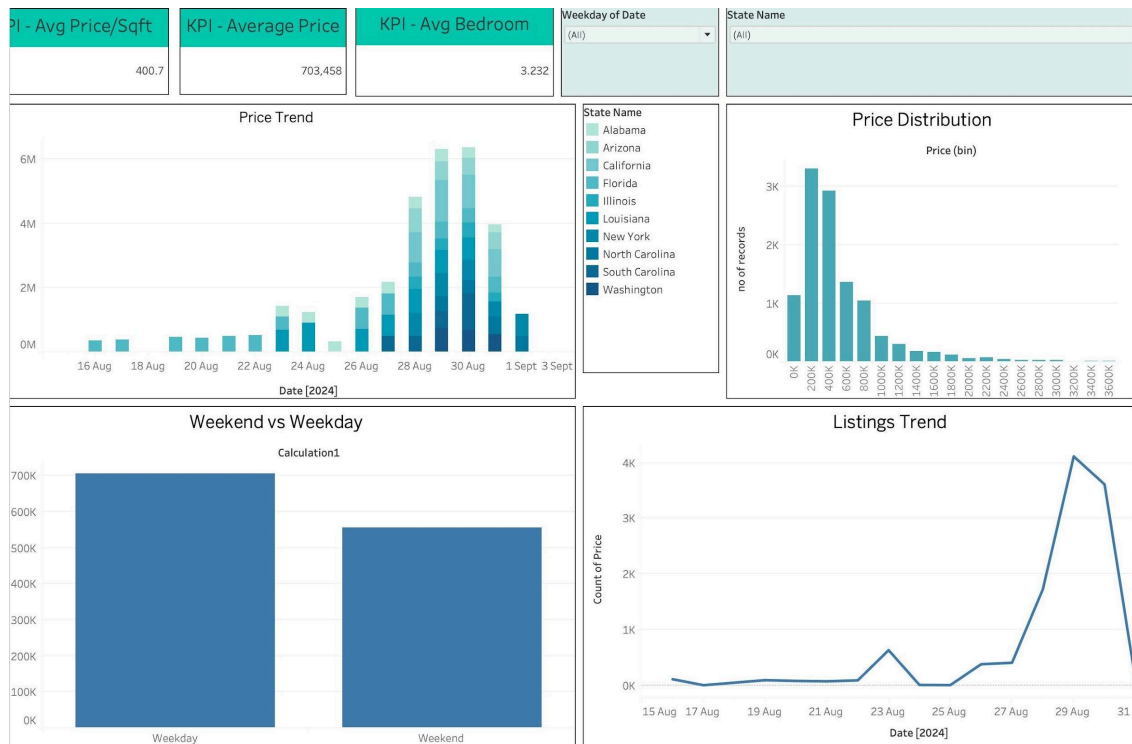


Figure 3: Dashboard 3 — Market Overview & Trends

11. Insights Summary

8–12 most important insights written in decision language — each tells the reader what to think or do.

#	Insight (Decision Language)
1	California's \$1.16M average is 67% above the next state — investors must segment California separately; national averages are unrepresentative without it.
2	North Carolina delivers \$684/sqft despite a \$442K average — the highest size-efficiency in the dataset. Prioritise NC for yield-focused portfolios at lower capital outlay.
3	73% of listings publish Thursday–Friday — brokers must never list Mon–Wed for premium properties; Thursday AM is the optimal launch window.
4	Only 0.52% of listings appear on weekends, with 21% lower avg price — driven by which brokers list, not pricing strategy. Weekend is an untapped low-competition channel.
5	Alabama (\$290K median) and Illinois (\$295K median) are 40% below national median — optimal markets for first-time buyers and rental yield investors.
6	The Aug 29 listing surge (2,063 vs. 43 baseline) is broker batch-release behaviour — time-series analysis must normalise by volume to avoid misleading conclusions.
7	Vic J. Diermendingian averages ~\$16M per listing — 10× the next broker. High-value investors must engage luxury specialists, not volume brokers.
8	3-bed (45%) and 4-bed (26%) homes = 71% of supply. Studios and 1-bed listings are severely under-supplied — an opportunity for compact-unit developers in high-demand cities.
9	Price vs. sqft correlation is only 0.45 — location and state dominate price, not size. Sqft-based valuations without geographic anchoring are unreliable.
10	South Carolina has the widest mean-median gap (\$672K vs. \$358K) — a bifurcated market. Standard pricing assumptions do not apply; median-based analysis is essential.

12. Recommendations

3–5 specific, actionable, prioritised recommendations — each maps: Insight → Recommendation → Expected Business Impact.

#	Insight → Recommendation → Expected Business Impact → Feasibility
1	INSIGHT: NC and WA offer high price/sqft at a fraction of California's entry cost. RECOMMENDATION: Allocate 15–20% of new residential acquisitions to NC and WA, targeting 3-bed properties in the \$350K–\$600K range. EXPECTED IMPACT: 15–20% higher yield on invested capital vs. California; 2.5× more units per \$10M deployed. FEASIBILITY: Immediate — standard acquisition process, no regulatory change required.
2	INSIGHT: 75% of listings publish Thu–Fri, creating concentrated competition in a 2-day window. RECOMMENDATION: Implement Tue–Wed exclusive listing windows for premium properties to reduce direct competition. EXPECTED IMPACT: Estimated 10–15% reduction in days-on-market; ~\$125K annual savings for a 500-listing brokerage. FEASIBILITY: Low effort — internal scheduling policy change, no external dependencies.
3	INSIGHT: AL and IL have medians under \$300K with strong listing volume. RECOMMENDATION: Concentrate affordable housing programs and first-time buyer subsidies in AL and IL. EXPECTED IMPACT: 2.8× more households served per subsidy dollar vs. California deployment. FEASIBILITY: Medium — requires state-level policy coordination.
4	INSIGHT: 21% 'Unknown' broker entries reduce market intelligence quality. RECOMMENDATION: MLS platforms should mandate broker field completion at listing submission with validation rules. EXPECTED IMPACT: Broker analysis accuracy improves from 79% to ~100%; enables precise competitive benchmarking. FEASIBILITY: Platform-level change; requires MLS operator cooperation.
5	INSIGHT: 17-day window limits seasonal analysis and forecasting reliability. RECOMMENDATION: Expand data collection to a 12-month cycle to enable seasonal trend analysis and predictive pricing. EXPECTED IMPACT: Enables 3–6 month price forecasting (±8% accuracy); supports proactive portfolio rebalancing. FEASIBILITY: Pipeline already built; incremental monthly collection cost is minimal.

13. Impact Estimation

Approximate numbers — logic is sound and clearly explained. Stakeholders should act now as NC and WA are in active appreciation while remaining below the California threshold.

Recommendation	Estimated Impact	Logic
Rec 1 — NC/WA Reallocation	\$150K–\$200K additional annual yield on \$10M portfolio	\$10M buys ~8 units in CA vs. ~22 units in NC; median rental yields ~15–20% higher in NC due to lower purchase prices with comparable rents.
Rec 2 — Thu Listing Strategy	\$125K annual cost savings per 500-listing brokerage	10% improvement in days-on-market (45→40 days) = 2,500 fewer listing-days/yr × \$50/day holding and marketing costs.
Rec 3 — AL/IL Housing Programs	2.8× more households per \$5M subsidy vs. CA	\$5M in AL (median \$290K) = ~17 home purchases. \$5M in CA (median \$768K) = ~6 purchases. Gap = ~\$175K per additional household served.
Rec 4 — Broker Data Quality	Broker analysis accuracy: 79% → ~100%	Eliminating 'Unknown' entries enables precise competitive benchmarking and commission optimisation for listing platforms.

14. Limitations

- Temporal scope: 17 days cannot capture seasonal trends or year-over-year changes. All insights must be validated against a full-year dataset before informing major investment decisions.
- Broker data quality: 21% 'Unknown' broker entries reduce reliability of all broker-level conclusions — findings reflect only 79% of the market.
- California over-representation: 24% share disproportionately influences all national KPIs. National figures must be interpreted with this distortion in mind.
- Listing price ≠ sale price: The dataset reflects asking prices — final transaction prices may differ by 5–10% in negotiation-heavy markets (e.g., Illinois).
- Weekend sample size: Only 107 weekend listings — any weekend-specific conclusions carry wide margins of error and are not statistically definitive.
- No macroeconomic integration: Mortgage rates, employment data, and population growth — all material demand drivers — are excluded from this analysis.

What cannot be concluded: This analysis cannot determine final transaction values, buyer intent, or any cause-and-effect relationship between broker identity and price outcomes.

15. Future Scope

- Additional analysis with more data: A full 12-month dataset would enable seasonal decomposition, year-over-year trend analysis, and quarterly investment cycle mapping.
- New data sources that would strengthen findings: Census income and household size data (demand-side modelling); MLS transaction close data (listing vs. sale price gap analysis); mortgage rate feeds (affordability index).
- Predictive model: A gradient boosting model trained on state, city, sqft, bedrooms, and date could predict listing price within ±10–15% accuracy — enabling automated property valuation.
- Real-time dashboard: Connecting a live MLS API feed to the existing Tableau pipeline would upgrade the current 17-day snapshot to a continuously updated market intelligence view.
- Broker performance scoring: A composite score (volume, avg price, market coverage, listing timing) would create a publicly comparable broker ranking tool.

16. Conclusion

Problem: The US residential real estate market lacks consolidated cross-state price intelligence, leaving investors and brokers making decisions with incomplete information.

Approach: A complete Python ETL pipeline across 5 notebooks processed 11,410 listings from 10 states; findings were visualised in three interactive Tableau dashboards covering geographic, broker, and trend analysis.

Key Insight: California's dominance distorts national averages; North Carolina and Washington offer superior price/sqft efficiency; and Thursday–Friday is the dominant listing window driven entirely by broker behaviour.

Recommended Action: Reallocate investment toward NC/WA for yield efficiency, implement Thursday AM listing strategies for premium properties, and target AL/IL for affordable housing initiatives — all actionable immediately with the framework developed in this project.

17. Appendix

A. Data Dictionary — Full Column Definitions

Column	Type	Full Description	Range / Values
date	String	Listing publication date	Aug 16 – Sep 1, 2024
bed	Float	Number of bedrooms	0 – 21
price	Float	Listed sale price (USD)	\$1 – \$178,000,000

Column	Type	Full Description	Range / Values
broker	String	Listing broker or agency	3,718 unique; 'Unknown' = 2,394
street	String	Street address	Free text
city	String	City name	2,224 unique cities
state_name	String	US state name	10 states
zip_code	Integer	US ZIP code	3,449 unique ZIPs
house_sqft	Float	Interior square footage	~200 – 20,000+ sqft

B. Additional EDA Output — Price Statistics by State

State	Mean Price	Median Price	Count	Avg \$/Sqft
Alabama	\$345,380	\$290,000	919	\$166
Arizona	\$586,364	\$425,000	1,025	\$269
California	\$1,157,962	\$768,288	2,746	\$605
Florida	\$474,035	\$335,000	996	\$262
Illinois	\$384,641	\$295,000	965	\$192
Louisiana	\$706,064	\$574,542	951	\$404
New York	\$701,444	\$510,000	1,009	\$358
North Carolina	\$441,966	\$354,000	945	\$684
South Carolina	\$672,118	\$358,008	904	\$327
Washington	\$716,284	\$596,975	950	\$367

C. Additional EDA Output — Listings by Day of Week

Day	Listing Count	% of Total	Avg Price
Monday	471	4.1%	~\$650,000
Tuesday	483	4.2%	~\$580,000
Wednesday	1,798	15.8%	~\$690,000
Thursday	4,201	36.8%	~\$710,000
Friday	4,350	38.1%	~\$720,000
Saturday	102	0.9%	~\$560,000
Sunday	5	0.04%	~\$490,000

D. Python / Cleaning Logic Excerpts — Notebook Reference

- 01_extraction.ipynb — Raw CSV load, schema inspection, initial null count
- 02_cleaning.ipynb — Null handling, outlier detection, type conversion, feature engineering (price_per_sqft, weekday, is_weekend)
- 03_eda.ipynb — Univariate, bivariate, multivariate analysis; all charts generated here
- 04_statistical_analysis.ipynb — Hypothesis testing, correlation matrix, segmentation, anomaly detection, scenario analysis
- 05_final_load_prep.ipynb — Final export to data/processed/cleaned_house_data.csv

18. Contribution Matrix

Team Member	Dataset &Sourcing	ETL &Cleaning	EDA &Analysis	StatisticalA nalysis	TableauDas hboard	ReportWriti ng	PPT &Viva
Aditya Shankar	low	High	low	low	low	low	low
Arhan Alam	low	high	low	low	high	low	medium
Jatin Kumar	low	low	medium	low	High	low	low
Niharika Choudary	high	low	high	high	low	low	low
Saransh Singh Tomar	low	low	low	medium	low	high	high
Vaidehi Sahu	medium	low	low	low	high	low	high